

University of Wah

Department of Computer Science



Submit to:

Ms. Wasifa Kanwal

Submit by:

Muhammad Talha

Course:

Information and Security Lab - CS-251L

Reg No:

UW-24-CS-BS-101

Section:

BS-CS-3rd-B

Task 1

DAY: _____

DATE: _____

Lab 11

Task 1:

Diffie-Hellman

Data :-

$$\text{prime number} = P = 101$$

$$\text{Primitive root} = g = 101$$

$$X_a = 8, X_b = 9$$

Solution

$$y = g^x \text{ mod } P$$

$$\Rightarrow y_a = (101)^8 \text{ mod } 101$$

$$y_a = 0$$

$$\Rightarrow y_b = g^{x_b} \text{ mod } P$$

$$y_b = (101)^9 \text{ mod } 101$$

$$y_b = 0$$

$$\Rightarrow K_A = y_b^{x_a} \text{ mod } P$$

$$= (0)^8 \text{ mod } 101$$

$$K_A = 0$$

$$K_B = y_a^{x_b} \text{ mod } P$$

$$= (0)^9 \text{ mod } 101$$

$$K_B = 0$$

$$K_A = K_B$$

Code Output

```
Public Key (e, n): (17, 3233)
Private Key (d, n): (2753, 3233)

Plain Text: Muhammad Talha
ASCII Values: [77, 117, 104, 97, 109, 109, 97, 100, 32, 84, 97, 108, 104, 97]

Encrypted Cipher Text: [3123, 2160, 2170, 1632, 2271, 2271, 1632, 1773, 1992, 2159, 1632, 745, 2170, 1632]

Decrypted ASCII Values: [77, 117, 104, 97, 109, 109, 97, 100, 32, 84, 97, 108, 104, 97]
Decrypted Text: Muhammad Talha
```

Task # 02

DATE: _____

RSA Algorithm :-

Data :-

$p = 61$
 $q = 53$

name = Muhammad Talha

Solution :-

$n = p \times q$
 $n = 61 \times 53$
 $n = 3233$

$\phi(n) = (p-1) \times (q-1)$
 $= (61-1) \times (53-1)$
 $= (60) \times (52)$
 $\phi(n) = 3120$

→ Public (e) where $\gcd(e, \phi(n)) = 1$
 $e = 2: \gcd(2, 3120) = 2$
 $e = 3: \gcd(3, 3120) = 3$
 $\vdots$
 $e = 7: \gcd(7, 3120) = 1$
 $e = 17$

→ Private Key (d)
 $(e \times d) \pmod{\phi(n)} = 1$
 $d = 2753$

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check:

$$7 \times 1783 = 12481$$

~~12481~~ Hence

$$12481 = (4 \times 3120) + 1$$

1783, ~~12481~~

$$(e, n) = (17, 3233)$$

$$(d, n) = (\frac{1}{17}, 3233)$$

2753

Text to Number:

M	w	h	a	m	m	a	d	(space)	T	a
77	117	104	97	109	109	97	100	32	84	97
l	h	a								
108	104	97								

Encryption -

$$e = 17$$

$$n = 3233$$

$$C = M^e \text{ mod } n$$

(M)

$$C = (77)^{17} \text{ mod } 3233$$

$$C = 1317$$

⇒ Same for all letters:

$$(w) \Rightarrow 221 \text{ (Ciphertext)}$$

$$(h) \Rightarrow 2170$$

$$(a) \Rightarrow 1632$$

$$(m) \Rightarrow 745$$

$$(m) \Rightarrow 745$$

$$(a) \Rightarrow 1632$$

$$(d) \Rightarrow 1773$$

DATE	
(space)	→ 1773
(T)	→ 1992
(a)	→ 2159
(l)	→ 1632
(h)	→ 3020
(m)	→ 2159 2159

Decryption

$$d = 2753$$

$$n = 3233$$

$$M = C^d \pmod{n}$$

$$C = (1317)^{2753} \pmod{3233}$$

$$77 \quad (M)$$

(Same for all letters)

$$(221) \Rightarrow ~~117~~ (u)$$

$$(2170) \Rightarrow 104 (h)$$

$$(1632) \Rightarrow 97 (a)$$

$$(745) \Rightarrow 109 (m)$$

$$(745) \Rightarrow 109 (m)$$

$$(1632) \Rightarrow 97 (a)$$

$$(1773) \Rightarrow 100 (d)$$

$$(1992) \Rightarrow 32 (\text{space})$$

$$(2159) \Rightarrow 84 (T)$$

$$(1632) \Rightarrow 97 (a)$$

$$(3020) \Rightarrow 108 (l)$$

$$(2170) \Rightarrow 104 (h)$$

$$(1632) \Rightarrow 97 (a)$$

Decrypt text:

Muhammad Talha

Code Output

```
Public Key (e, n): (5, 7990271)
Private Key (d, n): (1596269, 7990271)
Original Message: 123
Encrypted Message: 3332110
Decrypted Message: 123
PS: D:\University Data\3rd Semester\AI Lab\Final Project\DiagnoSightAI> ^C
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